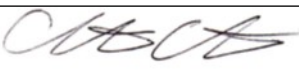


**Guideline Title: Pediatric Asystole**

**Guideline Number: GUID-3203-PED004**

|                                                                                              |                             |
|----------------------------------------------------------------------------------------------|-----------------------------|
| Issue date: 09/09/2014                                                                       | Replaces Dept. Policy:      |
| Revision dates: 4/30/24                                                                      | Developed by: Air Care Team |
| Approved by: Dr. Christopher Hunter, MD<br>Air Care Team Medical Director                    | Approved by:                |
| Signature:  | Signature:                  |
| Department Numbers: 3203                                                                     |                             |

- I. **PURPOSE:** To ensure prompt identification and intervention to restore a stable cardiac rhythm.
- II. **PATHOPHYSIOLOGY:** Asystole is the absence of electrical activity in the heart. The approach to asystole resuscitation is essentially the same as pulseless electrical activity (PEA). Despite dismal survival rates after out of hospital cardiac arrest, pediatric patients in asystole or PEA have higher survival rates and better neurologic outcomes than adults. Thus, aggressive resuscitation for longer periods of time are warranted.
- III. **DEFINITIONS:**
  - A. AHA – American Heart Association
  - B. CPR – Cardiopulmonary resuscitation
- IV. **DEPARTMENT GUIDELINE:**
  - A. Initiate CPR per AHA guideline with compression rate of 100-120/min.
  - B. Apply cardiac monitor. Confirm asystole in two leads. Asystole should not be defibrillated.
    - i. Perform rhythm analysis every 2 minutes between cycles of CPR.
  - C. Obtain or confirm IV/IO access.
  - D. Manage airway and ensure adequate oxygenation and ventilation with positive pressure ventilation.
    - i. Initially administer O2 via BVM
    - ii. Secure airway and ventilate with 100% oxygen. Intubate, if necessary, per *GUID3203-PR020 Endotracheal Intubation*.
    - iii. Ventilate with 1 breath every 2 to 3 seconds (a rate of about 20-30 breaths per minute). Avoid hyperventilation.
    - iv. Apply ETCO2 monitor if advanced airway present.
  - E. Administer
    - i. **Epinephrine (1:10,000) 0.01 mg/kg (0.1 ml/kg) IV/IO**, max dose 1 mg.
    - ii. **Epinephrine (1:1,000) 0.1 mg/kg (0.1 ml/kg) via ETT**, max dose 1 mg
    - iii. Re-dose every 3-5 minutes
  - F. Administer 0.9% NS or Lactated Ringers 20 ml/kg IV
  - G. Consider underlying causes (5 H's and T's): Hypovolemia, Hypoxia, H+ ions – Acidosis, Hyper/Hypokalemia, Hypothermia, Hypoglycemia, Toxic Ingestion, cardiac tamponade, Tension Pneumothorax, Thrombus – AMI/PE, Trauma. Also: congenital heart disease and pulmonary hypertension.
  - H. If cardiac arrest due to Trauma

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- i. Anticipate airway obstruction by dental fragments, blood, or other debris
- ii. Minimize cervical spine motion
- iii. Control external bleeding with pressure or per *GUID3203-PR006 Tourniquet Therapy* or *GUID-PR028 Hemostatic Dressing - QuickClot Gauze*
- iv. Perform unilateral or bilateral needle decompressions for suspected tension pneumothoraxes per *GUID3203-PR017 Needle Thoracostomy*. Pneumothorax is rare in the pediatric population.
- v. Transport infants and children with multisystem trauma to the closest pediatric trauma center
- vi. Initiate volume resuscitation with blood products per *GUID3203-PR025 - Emergency Blood Product Administration*
- vii. Non-accidental trauma should be considered for any infant with unanticipated, unexplained cardiac arrest
- I. For deterioration during transport after ROSC, consider mechanical problems with the airway, ventilation, and/or blood volume. Dysrhythmias are rare, except as an agonal event. Consider: extubation, plugged endotracheal tube, right main stem bronchus intubation, distended stomach causing impaired lung expansion, pneumothorax, inadequate minute ventilation, ventilator malfunction, disconnected or depleted oxygen supply, hypovolemic shock, or agitation.
- J. Report Broselow color and/or estimated weight to receiving center prior to arrival
- K. Out of hospital cessation of resuscitation efforts should not be considered in pediatric patients without contacting medical control.
- L. Always consider the latest recommendations from the American Heart Association.

V. **DOCUMENTATION:** EMS Charts

VI. **REFERENCES:**

- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.